

SAT Math Mastery · Week 2B

Linear Modeling & Inequality Regions

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics thresholds and tiered rates · mixtures · whole-number constraints · linear inequalities in context · shaded regions and inequality systems · feasible points · triangle inequality · unit shifts · meaning of parameters

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Build a linear function with a flat part, a threshold and a rate: $C = F + r(n - k)$ for $n \geq k$.
- S2.** Handle tiers, discounts, tax and percent factors in the correct order.
- S3.** Translate phrases (at most, at least, between, no more than) into inequalities and solve them.
- S4.** Round whole-number answers in the feasible direction, and check the boundary value.
- S5.** Model two-item mixtures and percent mixtures with a count equation and a value equation.
- S6.** Read a shaded region: find the boundary line, decide solid or dashed, test a point for the direction.
- S7.** Decide whether an ordered pair satisfies a system of inequalities (every inequality, strict vs. non-strict).
- S8.** Use the triangle inequality, unit shifts through a linear conversion, and the meaning of slope and constants in context.

2 · REQUIRED FACTS

- “At most” means $\leq$; “at least” means $\geq$; “fewer than” means $<$; “between a and b , inclusive” means $a \leq x \leq b$.
- Multiplying or dividing an inequality by a negative number reverses the sign.
- A solid boundary line includes its points ($\leq, \geq$); a dashed boundary line does not ($<, >$).
- A point is a solution of a system of inequalities only if it satisfies *every* inequality.
- The graph of $y \geq mx + b$ is shaded above the line, $y \leq mx + b$ below it; always confirm with a test point such as $(0, 0)$ when it is not on the line.
- For whole-number quantities (items, hours, people) the answer is an integer on the feasible side of the boundary.
- Triangle inequality: each side is shorter than the sum of the other two; the third side x satisfies $|a - b| < x < a + b$.
- A change Δx in the input of $y = mx + b$ changes the output by $m \Delta x$; the constant b cancels.

3 · REQUIRED FORMULAS

$C(n) = F + r(n - k)$ ($n \geq k$), total with tax rate p : $(1 + p) \cdot \text{pre-tax total}$, after $p\%$ off: $(1 - p) \cdot \text{price}$

mixture: $x + y = T$, $c_1x + c_2y = cT$, $|a - b| < x < a + b$, $\Delta y = m \Delta x$

feasible region: all inequalities hold together; corner points come from intersecting two boundary lines

4 · MINI THEORY SHEET

Phrase to inequality.

Words	Symbol and round-off rule
at most, no more than, cannot exceed	$\leq$; for a maximum count, round <i>down</i>
at least, no fewer than, minimum of	$\geq$; for a minimum count, round <i>up</i>
more than, exceeds, less than	$>$ or $<$; test the boundary value itself
between a and b	$a < x < b$ (inclusive: $a \leq x \leq b$)

Thresholds and tiers. Write the cost of each tier separately and add them: first k units at the flat price, then rate $\times$ (units $- k$). Apply a discount only to the units it covers; apply tax to the final subtotal.

Mixtures. Use a count equation and a value equation. For concentrations: (amount) $\times$ (concentration) summed over the parts equals (total amount) $\times$ (final concentration). Adding pure water or a pure substance means that part has concentration 0% or 100%.

Regions. Find two lattice points on the boundary, write the line, then test a point not on it. Solid line $\Rightarrow \leq$ or $\geq$; dashed line $\Rightarrow <$ or $>$. For an overlap region, repeat for each boundary.

Feasible points. Test a candidate in every constraint. The correct choice satisfies all of them; each wrong choice usually breaks exactly one.

Meaning in context. In $y = mx + b$: m is the change in y per 1 unit of x , b the value at $x = 0$. A product such as $0.85 \cdot 6$ combines both a discount factor and a rate: ask “per what?”.

Speed shortcuts.

Situation	Shortcut
maximum whole-number count	solve the inequality, then take the floor and recheck the cost
extra units beyond a threshold	charge only $(n - k)$ units at the new rate
percent change and tax together	multiply the factors: 0.65×1.08
two-item mixture, one unknown asked	total value $-$ (all cheaper item) $\div$ price gap
shaded region	line through two lattice points, then test the origin
integer points in a system	plug in the endpoints of each inequality, strict bounds excluded
triangle with sides a, b, x	$ a - b < x < a + b$; count integers carefully
degrees F vs. C (or K)	$\Delta F = 1.8 \Delta C$; ignore the $+32$

5 · GUIDED EXAMPLES

Example 1 — Threshold and whole-number rounding

A taxi charges \$4 for the first 2 miles and \$1.80 for each additional mile (whole miles only). What is the greatest trip, in miles, that costs no more than \$20?

For $m \geq 2$: $C = 4 + 1.80(m - 2) \leq 20 \Rightarrow m - 2 \leq \frac{16}{1.8} \approx 8.89$. Whole miles: round *down* to $m - 2 = 8$, so $m = 10$. Check: $m = 10$ costs \$18.40; $m = 11$ costs \$20.20 $>$ \$20.

Example 2 — From a shaded region to an inequality

The boundary is a dashed line through $(0, 4)$ and $(2, 0)$; the side containing the origin is shaded. Write the inequality.

Slope $= \frac{0 - 4}{2 - 0} = -2$, so $y = -2x + 4$, that is $2x + y = 4$. Test $(0, 0)$: $0 < 4$ is true, so the shaded side is $2x + y < 4$ (dashed line, so strict).

Example 3 — Percent mixture

How many liters of a 10% solution and of a 40% solution make 30 liters of a 20% solution?

$x + y = 30$ and $0.10x + 0.40y = 0.20(30) = 6$. Substitute $x = 30 - y$: $3 + 0.30y = 6 \Rightarrow y = 10$ liters of the 40% solution and $x = 20$ liters of the 10% solution.

6 · PRACTICE SET — 37 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED
A streaming service charges a one-time sign-up fee of \$12 plus \$9 per month. Priya wants to spend no more than \$100 in total. Which inequality can be used to find m , the number of months she can subscribe?

(A) $9(12 + m) \leq 100$

(B) $12 + 9m \leq 100$ ✓

(C) $12m + 9 \leq 100$

(D) $12 + 9m \geq 100$

- Given: a one-time sign-up fee of \$12, a charge of \$9 per month, and a total of no more than \$100. Let m = the number of months Priya subscribes. We want an inequality in m .
- Method: build a linear model for the total cost, then turn the words 'no more than' into an inequality sign.
- The monthly charges: each month costs \$9, so m months cost $9 \cdot m = 9m$ dollars.
- The sign-up fee is paid only once, so it is added and not multiplied: total cost = $12 + 9m$.
- 'No more than \$100' means the total may equal 100 or be smaller, so we use $\leq$: $12 + 9m \leq 100$.
- Check with $m = 8$: $12 + 9(8) = 12 + 72 = 84$, and $84 \leq 100$ is true. Check with $m = 10$: $12 + 9(10) = 12 + 90 = 102$, and $102 \leq 100$ is false. The inequality behaves as the story says.
- Compare with the choices: option B is $12 + 9m \leq 100$, which matches. Option A expands to $108 + 9m \leq 100$, which multiplies the fee by 9; option C charges \$12 per month; option D uses $\geq$.
- So the answer is B, $12 + 9m \leq 100$.

Answer: (B)

2. EASY · NO CALCULATOR NEEDED
Two sides of a triangle have lengths 7 and 12. Which of the following could be the length of the third side?

(A) 21

(B) 5

(C) 11 ✓

(D) 19

- Given: two sides of a triangle are 7 and 12. Let x = the third side. We want the option that x could equal.
- Method: the triangle inequality says the sum of any two sides must be greater than the third side.
- Condition 1: $7 + 12 > x$. Adding gives $19 > x$, so $x < 19$.
- Condition 2: $7 + x > 12$. Subtract 7 from both sides: $x > 5$.
- Condition 3: $12 + x > 7$. Subtract 12 from both sides: $x > -5$. This is true for every positive length, so it adds nothing new.
- Combine the conditions: $5 < x < 19$. Both ends are strict, so x cannot equal 5 or 19.
- Test the choices: $21 < 19$ is false; $5 > 5$ is false; $5 < 11 < 19$ is true; $19 < 19$ is false.
- Check 11 in all three sums: $7 + 11 = 18 > 12$, $12 + 11 = 23 > 7$, $7 + 12 = 19 > 11$. All are true. So the answer is C, 11.

Answer: (C)

3. EASY · NO CALCULATOR NEEDED
The temperature in degrees Fahrenheit, F , is related to the temperature in degrees Celsius, C , by $F = 1.8C + 32$. If the temperature rises by 10 degrees Celsius, by how many degrees Fahrenheit does it rise?

- Given: $F = 1.8C + 32$, and C increases by 10. We want the increase in F , which is (new F) – (old F).
- Method: write F before and after the change and subtract. The slope 1.8 is the change in F for each 1 degree change in C .
- Let the starting Celsius temperature be C . Then the new Celsius temperature is $C + 10$.
- Old Fahrenheit temperature: $F_1 = 1.8C + 32$.
- New Fahrenheit temperature: $F_2 = 1.8(C + 10) + 32$. Distribute the 1.8: $F_2 = 1.8C + 18 + 32$.
- Subtract: $F_2 - F_1 = (1.8C + 18 + 32) - (1.8C + 32)$. The $1.8C$ terms cancel ($1.8C - 1.8C = 0$) and the 32 terms cancel ($32 - 32 = 0$), leaving 18.
- Check with $C = 0$: $F = 1.8(0) + 32 = 32$. With $C = 10$: $F = 1.8(10) + 32 = 18 + 32 = 50$. The rise is $50 - 32 = 18$.
- So the answer is 18 degrees Fahrenheit.

Answer: 18

4.

EASY · NO CALCULATOR NEEDED

$$y > 2x - 1$$

$$y < -x + 5$$

Which of the following ordered pairs (x, y) is a solution to the system of inequalities above?

(A) $(1, 2)$ ✓(B) $(2, 3)$ (C) $(-1, 6)$ (D) $(4, -2)$

- Given: the system $y > 2x - 1$ and $y < -x + 5$. We want the ordered pair (x, y) that makes both inequalities true.
- Method: test each choice by substituting its x and y into both inequalities. A solution must make both true.
- Choice A, $(1, 2)$, first inequality: $2x - 1 = 2(1) - 1 = 2 - 1 = 1$, and $2 > 1$ is true.
- Choice A, second inequality: $-x + 5 = -1 + 5 = 4$, and $2 < 4$ is true. Both are true, so $(1, 2)$ is a solution.
- Choice B, $(2, 3)$, first inequality: $2(2) - 1 = 3$, and $3 > 3$ is false because 3 equals 3 and the inequality is strict. B fails.
- Choice C, $(-1, 6)$, second inequality: $-(-1) + 5 = 1 + 5 = 6$, and $6 < 6$ is false. C fails.
- Choice D, $(4, -2)$, first inequality: $2(4) - 1 = 7$, and $-2 > 7$ is false. D fails.
- Only $(1, 2)$ satisfies both inequalities. So the answer is A, $(1, 2)$.

Answer: (A)

5.

EASY · CALCULATOR

A parking garage charges \$6 for the first 2 hours and \$2.50 for each additional hour. What is the cost, in dollars, of parking for 7 hours?

- Given: the first 2 hours cost \$6, each additional hour costs \$2.50, and the car is parked for 7 hours. We want the total cost in dollars.
- Method: split the parking time into the part covered by the flat \$6 and the additional hours.
- Additional hours = $7 - 2 = 5$ hours.
- Cost of the additional hours: $5 \times 2.50 = 12.50$ dollars.
- Total cost = $6 + 12.50 = 18.50$ dollars.
- Check hour by hour: hours 1 and 2 cost \$6 together; hours 3, 4, 5, 6, 7 are 5 hours at \$2.50, which is \$12.50. The total is $6 + 12.5 = 18.5$.
- So the answer is 18.5 dollars (that is, \$18.50).

Answer: 18.5

6.

MEDIUM · NO CALCULATOR NEEDED

A delivery service charges a flat \$8 for the first 3 kilometers and \$1.50 for each kilometer after the first 3. Which equation gives the cost C , in dollars, of a trip of d kilometers, where $d \geq 3$?

(A) $C = 1.50(d - 3)$ (B) $C = 8 + 1.50(d + 3)$ (C) $C = 8 + 1.50d$ (D) $C = 8 + 1.50(d - 3)$ ✓

- Given: a flat \$8 covers the first 3 km, each kilometer after the first 3 costs \$1.50, and $d \geq 3$. Let C = the cost in dollars and d = the distance in kilometers. We want an equation for C .
- Method: build a linear model: flat fee plus rate times the extra distance.
- The distance charged at \$1.50 is only the part beyond 3 km. For example, $d = 7$ gives $7 - 3 = 4$ extra kilometers. In general, extra distance = $d - 3$.
- Charge for the extra distance: $1.50(d - 3)$.
- Add the flat fee: $C = 8 + 1.50(d - 3)$.
- For reference, distribute: $C = 8 + 1.50d - 4.50$. Combine the constants: $C = 1.50d + 3.50$.
- Check $d = 3$: $C = 8 + 1.50(0) = 8$, which is the flat fee alone. Check $d = 7$: $C = 8 + 1.50(4) = 8 + 6 = 14$, and by hand \$8 plus 4 km at \$1.50 is $8 + 6 = 14$. Both agree.
- Compare the choices: option D is $C = 8 + 1.50(d - 3)$, which matches. Option A has no \$8 fee, option B uses $d + 3$, and option C at $d = 3$ gives $8 + 4.50 = 12.50$, not 8.
- So the answer is D, $C = 8 + 1.50(d - 3)$.

Answer: (D)

7.

MEDIUM · CALCULATOR

An elevator can carry at most 1,200 pounds. An operator who weighs 160 pounds is in the elevator, and each box loaded weighs 85 pounds. What is the greatest number of boxes that can be loaded?

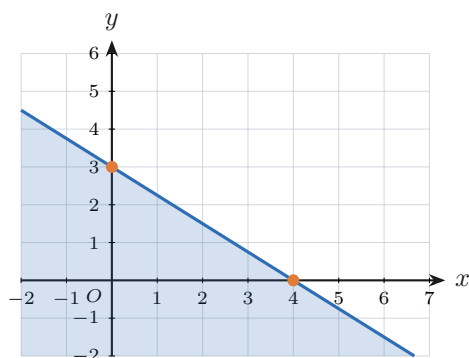
- Given: the limit is 1200 pounds, the operator weighs 160 pounds, and each box weighs 85 pounds. Let b = the number of boxes. We want the greatest whole number b .
- Method: write an inequality for the total weight, solve it, and round down because boxes come in whole numbers.
- Total weight is operator plus boxes, and it must not exceed the limit: $160 + 85b \leq 1200$.
- Subtract 160 from both sides: $85b \leq 1040$, because $1200 - 160 = 1040$.
- Divide both sides by 85. Since 85 is positive, the inequality sign stays the same: $b \leq \frac{1040}{85}$.
- Compute the division: $85 \times 12 = 1020$ and $1040 - 1020 = 20$, so $\frac{1040}{85} = 12\frac{20}{85} \approx 12.24$.
- So $b \leq 12.24$. The greatest whole number not above 12.24 is 12.
- Check $b = 12$: $160 + 85(12) = 160 + 1020 = 1180$, and $1180 \leq 1200$ is true. Check $b = 13$: $160 + 85(13) = 160 + 1105 = 1265$, and $1265 \leq 1200$ is false.
- So the answer is 12 boxes.

Answer: 12

8.

MEDIUM · NO CALCULATOR NEEDED

The shaded region in the xy -plane is the graph of a linear inequality, and the boundary line is solid. Which inequality represents the shaded region?



(A) $4x + 3y \leq 12$

(B) $3x + 4y < 12$

(C) $3x + 4y \leq 12$ ✓

(D) $3x + 4y \geq 12$

- Given (from the figure): the boundary line is solid and passes through $(0, 3)$ and $(4, 0)$, and the origin lies in the shaded region. We want the inequality for the shaded region.
- Method: find the equation of the boundary line, then test a point to decide which side is shaded, then use the line style to choose between strict and non-strict.
- Slope of the line: $m = \frac{0 - 3}{4 - 0} = -\frac{3}{4}$.
- The y -intercept is 3, so the line is $y = -\frac{3}{4}x + 3$.
- Multiply both sides by 4 to clear the fraction: $4y = -3x + 12$.
- Add $3x$ to both sides: $3x + 4y = 12$.
- Check both points on the line: $(0, 3)$ gives $3(0) + 4(3) = 12$, and $(4, 0)$ gives $3(4) + 4(0) = 12$. Both work.
- Test the origin, which is shaded: $3(0) + 4(0) = 0$, and $0 < 12$. So the shaded side is where $3x + 4y$ is less than 12.
- The boundary is solid, so points on the line are included. We therefore use the non-strict sign: $3x + 4y \leq 12$.
- Compare the choices: option C is $3x + 4y \leq 12$, which matches. Option A has the line $4x + 3y = 12$, which gives 9 at $(0, 3)$, not 12. Option B is strict, which needs a dashed line. Option D would need $0 \geq 12$ at the origin, which is false. So the answer is C.

Answer: (C)

9.

MEDIUM · CALCULATOR

A shop mixes coffee beans that cost \$8 per pound with coffee beans that cost \$12 per pound to make 20 pounds of a blend that costs \$9.50 per pound. How many pounds of the \$12 beans are in the blend?

- Given: beans at \$8 per pound and \$12 per pound make 20 pounds of blend at \$9.50 per pound. Let b = pounds of \$12 beans. Then the \$8 beans weigh $20 - b$ pounds. We want b .
- Method: the cost of the \$8 beans plus the cost of the \$12 beans must equal the total cost of the blend.
- Total cost of the blend: $20 \times 9.50 = 190$ dollars.
- Write the equation: $8(20 - b) + 12b = 190$.
- Distribute the 8: $160 - 8b + 12b = 190$.
- Combine like terms ($-8b + 12b = 4b$): $160 + 4b = 190$.
- Subtract 160 from both sides: $4b = 30$.
- Divide both sides by 4: $b = 7.5$.
- Check: the \$8 beans weigh $20 - 7.5 = 12.5$ pounds and cost $8(12.5) = 100$. The \$12 beans cost $12(7.5) = 90$. Total $= 100 + 90 = 190$, which matches.
- So the answer is 7.5 pounds.

Answer: 7.5

10.

MEDIUM · NO CALCULATOR NEEDED

A triangle has side lengths 6, 10 and x , where x is an integer. What is the sum of all possible values of x ?

- Given: a triangle with sides 6, 10 and x , where x is an integer. We want the sum of all possible values of x .
- Method: use the triangle inequality to find the range of x , list the integers in it, and add them.
- Condition 1: $6 + 10 > x$, so $16 > x$, that is $x < 16$.
- Condition 2: $6 + x > 10$. Subtract 6 from both sides: $x > 4$.
- Condition 3: $10 + x > 6$. Subtract 10 from both sides: $x > -4$. This is true for every positive length.
- Combine: $4 < x < 16$, with both ends strict.
- The integers are 5, 6, 7, ..., 15. The count is $15 - 5 + 1 = 11$.
- Sum: the average of the first and last is $\frac{5 + 15}{2} = 10$, so the sum is $11 \times 10 = 110$.
- Check by pairing: $(5 + 15) + (6 + 14) + (7 + 13) + (8 + 12) + (9 + 11) + 10 = 5(20) + 10 = 110$.
- So the answer is 110.

Answer: 110

11.

MEDIUM · NO CALCULATOR NEEDED

A museum membership costs \$40 plus \$6 per visit, and members receive 15% off their total. The member's total cost, in dollars, for v visits is $C = 0.85(40 + 6v)$. Which is the best interpretation of the value $0.85 \cdot 6$ in this context?

(A) The percent discount on each visit

(B) The cost of one visit after the discount ✓

(C) The amount saved on each visit

(D) The cost of 6 visits after the discount

- Given: $C = 0.85(40 + 6v)$, where \$40 is the membership, \$6 is the price per visit, and 15% is taken off. We want the meaning of $0.85 \cdot 6$.
- Method: expand the formula into the form $C = \text{constant} + (\text{rate})v$ and read what the coefficient of v means.
- Distribute 0.85 over both terms: $C = 0.85(40) + 0.85(6v)$.
- Compute: $0.85 \times 40 = 34$ and $0.85 \times 6 = 5.1$. So $C = 34 + 5.1v$.
- The coefficient of v is $5.1 = 0.85 \cdot 6$. It is the number of dollars C increases for each additional visit.
- Why 0.85: a 15% discount means the member pays $100\% - 15\% = 85\% = 0.85$ of the price. Since \$6 is the price of one visit, $0.85 \cdot 6$ is what one visit costs after the discount.
- Check with numbers: $C(0) = 34$ and $C(1) = 34 + 5.1 = 39.1$. The difference is 5.1 dollars for one more visit.
- Compare the choices: $0.85 \cdot 6$ is not a percent (A); the amount saved per visit would be $0.15 \cdot 6 = 0.9$ (C); the cost of 6 visits would be $6 \times 5.1 = 30.6$ (D). Option B, the cost of one visit after the discount, matches.
- So the answer is B.

Answer: (B)

12.

MEDIUM · NO CALCULATOR NEEDED

Maria earns p dollars per month and saves 15% of it. She wants her monthly savings to be at least \$270 and at most \$450. Which inequality gives all possible values of p ?

(A) $317.6 \leq p \leq 529.4$

(B) $40.5 \leq p \leq 67.5$

(C) $270 \leq p \leq 450$

(D) $1800 \leq p \leq 3000$ ✓

- Given: Maria saves 15% of her monthly earnings p , and her savings must be at least \$270 and at most \$450. Let p = monthly earnings in dollars. We want the possible values of p .
- Savings = 15% of $p = 0.15p$.
- Method: write a compound inequality for the savings, then isolate p by dividing every part by 0.15.
- 'At least 270' means ≥ 270 and 'at most 450' means ≤ 450 . So $270 \leq 0.15p \leq 450$.
- Divide every part by 0.15. Since 0.15 is positive, the signs stay the same: $\frac{270}{0.15} \leq p \leq \frac{450}{0.15}$.
- Compute: $\frac{270}{0.15} = \frac{27000}{15} = 1800$ and $\frac{450}{0.15} = \frac{45000}{15} = 3000$.
- So $1800 \leq p \leq 3000$.
- Check the ends: $0.15(1800) = 270$ and $0.15(3000) = 450$. A middle value, $p = 2000$, gives $0.15(2000) = 300$, which is between 270 and 450.
- Option D is $1800 \leq p \leq 3000$. So the answer is D.

Answer: (D)**13.**

MEDIUM · NO CALCULATOR NEEDED

A student buys n notebooks at \$2.50 each and p pens at \$1.25 each. She spends at most \$30, buys at least 4 notebooks, and buys at least twice as many pens as notebooks. Which of the following pairs (n, p) is possible?

(A) $(5, 10)$ ✓

(B) $(3, 8)$

(C) $(5, 9)$

(D) $(6, 16)$

- Given: n notebooks at \$2.50 each and p pens at \$1.25 each. She spends at most \$30, buys at least 4 notebooks, and buys at least twice as many pens as notebooks. We want the possible pair (n, p) .
- Write the three conditions: (i) $n \geq 4$; (ii) $p \geq 2n$; (iii) $2.50n + 1.25p \leq 30$.
- Method: test each choice against all three conditions. The correct choice must satisfy every one.
- Choice A, $(5, 10)$: (i) $5 \geq 4$ true. (ii) $2(5) = 10$ and $10 \geq 10$ true. (iii) $2.50(5) = 12.50$ and $1.25(10) = 12.50$, so the cost is $25 \leq 30$, true. All three hold.
- Choice B, $(3, 8)$: (i) $3 \geq 4$ is false. B fails.
- Choice C, $(5, 9)$: (ii) $2(5) = 10$ and $9 \geq 10$ is false. C fails.
- Choice D, $(6, 16)$: (i) and (ii) hold ($6 \geq 4$ and $16 \geq 12$), but (iii) gives $2.50(6) + 1.25(16) = 15 + 20 = 35$, and $35 \leq 30$ is false. D fails.
- Only $(5, 10)$ satisfies all three conditions. So the answer is A, $(5, 10)$.

Answer: (A)

14.

HARD · NO CALCULATOR NEEDED

A mobile data plan costs \$20 per month and includes the first 5 gigabytes (GB). Each of the next 10 GB costs \$3 per GB, and each GB beyond the first 15 GB costs \$6 per GB. Data is billed in whole GB. What is the greatest amount of data, in GB, that can be used in a month for a total cost of at most \$100?

(A) 31

(B) 23 ✓

(C) 24

(D) 28

1. Given: \$20 per month covers the first 5 GB; each GB from 6 to 15 costs \$3; each GB beyond 15 costs \$6; billing is in whole GB; the total must be at most \$100. Let g = total GB used. We want the greatest g .
2. Method: a piecewise cost. First find the cost of using exactly 15 GB, then see how many \$6 gigabytes the leftover budget buys.
3. GB 6 through 15 is 10 GB at \$3: $10 \times 3 = 30$ dollars.
4. Cost through 15 GB: $20 + 30 = 50$ dollars. This is under \$100, so more than 15 GB can be used.
5. For $g > 15$ there are $g - 15$ extra GB at \$6 each, so the total is $50 + 6(g - 15)$. Require $50 + 6(g - 15) \leq 100$.
6. Subtract 50 from both sides: $6(g - 15) \leq 50$.
7. Divide both sides by 6 (positive, so the sign stays): $g - 15 \leq \frac{50}{6} = 8.33\dots$
8. Billing is in whole GB, so $g - 15$ must be a whole number not above 8.33. The greatest is $g - 15 = 8$.
9. Add 15 to both sides: $g = 23$.
10. Check $g = 23$: $50 + 6(8) = 50 + 48 = 98$, and $98 \leq 100$ is true. Check $g = 24$: $50 + 6(9) = 50 + 54 = 104$, and $104 \leq 100$ is false.
11. Option B is 23. So the answer is B, 23 GB.

Answer: (B)

15.

HARD · CALCULATOR

A store sells jackets for \$60 each. A customer pays full price for the first jacket, and each additional jacket is 35% off. An 8% sales tax is then added to the total. What is the greatest number of jackets a customer can buy with \$400?

(A) 6

(B) 7

(C) 8 ✓

(D) 9

1. Given: jackets cost \$60; the first is full price and each additional one is 35% off; an 8% tax is added to the total; the budget is \$400. Let n = the number of jackets. We want the greatest n .
2. Method: build the total cost as a function of n , then solve the inequality (total) ≤ 400 .
3. A 35% discount means paying 65% of the price: $60(0.65) = 39$ dollars per additional jacket.
4. With n jackets, one is full price and $n - 1$ are discounted, so the cost before tax is $60 + 39(n - 1)$.
5. Tax of 8% multiplies the total by 1.08, so we need $1.08[60 + 39(n - 1)] \leq 400$.
6. Divide both sides by 1.08: $60 + 39(n - 1) \leq \frac{400}{1.08} \approx 370.37$.
7. Subtract 60 from both sides: $39(n - 1) \leq 310.37$.
8. Divide both sides by 39: $n - 1 \leq 7.96$ (since $39 \times 7 = 273$ and $39 \times 8 = 312$, which is more than 310.37).
9. Jackets are whole numbers, so $n - 1 \leq 7$.
10. Add 1 to both sides: $n \leq 8$.
11. Check $n = 8$: before tax $60 + 39(7) = 60 + 273 = 333$; with tax $333 \times 1.08 = 359.64 \leq 400$, true. Check $n = 9$: before tax $60 + 39(8) = 372$; with tax $372 \times 1.08 = 401.76$, and $401.76 \leq 400$ is false.
12. Option C is 8. So the answer is C, 8.

Answer: (C)

16.

HARD · NO CALCULATOR NEEDED

A club has \$900 to spend on T-shirts and hoodies. Each T-shirt costs \$9 and each hoodie costs \$24. The club wants to buy at least 50 items in all and at least twice as many T-shirts as hoodies. What is the greatest number of hoodies the club can buy?

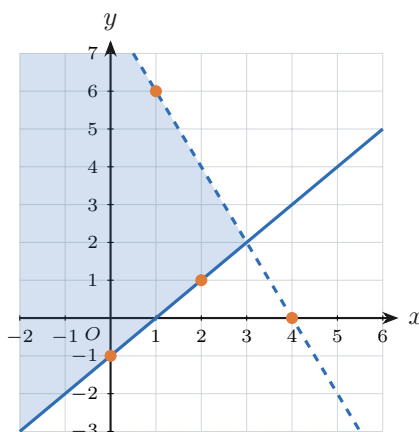
- Given: budget \$900; a T-shirt costs \$9 and a hoodie costs \$24; at least 50 items; at least twice as many T-shirts as hoodies. Let x = number of T-shirts and y = number of hoodies (whole numbers). We want the greatest y .
- The conditions are: $9x + 24y \leq 900$, $x + y \geq 50$, and $x \geq 2y$.
- Method: to afford as many hoodies as possible, spend as little as possible on T-shirts, so use the fewest T-shirts the rules allow. The smallest allowed x is the larger of $2y$ and $50 - y$.
- Compare $2y$ and $50 - y$: $2y \geq 50 - y$ when $3y \geq 50$, that is $y \geq 16.67$. So for $y \geq 17$ the smallest allowed x is $2y$.
- With $x = 2y$ the cost is $9(2y) + 24y = 18y + 24y = 42y$. We need $42y \leq 900$.
- Divide both sides by 42: $y \leq \frac{900}{42} = 21.43 \dots$ (since $42 \times 21 = 882$ and $42 \times 22 = 924$). So $y \leq 21$.
- Test $y = 21$: the smallest x is $\max(2 \cdot 21, 50 - 21) = \max(42, 29) = 42$. Cost: $9(42) = 378$ and $24(21) = 504$, so $378 + 504 = 882 \leq 900$ is true.
- Also for $y = 21$, $x = 42$: $x + y = 63 \geq 50$ is true and $x = 42 \geq 2(21) = 42$ is true. So 21 hoodies is possible.
- Test $y = 22$: the smallest x is $\max(44, 28) = 44$. Cost: $9(44) = 396$ and $24(22) = 528$, so $396 + 528 = 924$, and $924 \leq 900$ is false. Any larger x costs even more, so 22 is not possible.
- Larger y costs more still, so no number of hoodies above 21 works. So the answer is 21.

Answer: 21

17.

HARD · NO CALCULATOR NEEDED

The shaded region in the xy -plane is the solution set of a system of two linear inequalities. One boundary line is solid and the other is dashed. Which system of inequalities has the shaded region as its solution set?

(A) $y \geq x - 1$ and $y < -2x + 8$ ✓(B) $y \leq x - 1$ and $y < -2x + 8$ (C) $y \geq x - 1$ and $y > -2x + 8$ (D) $y > x - 1$ and $y \leq -2x + 8$

- Given (from the figure): the solid boundary passes through (0, -1) and (2, 1); the dashed boundary passes through (1, 6) and (4, 0); the point (0, 5) is in the shaded region. We want the system for the shaded region.
- Method: find each line's equation, test (0, 5) to find which side is shaded, and use solid for $\leq, \geq$ and dashed for $<, >$.
- Solid line slope: $\frac{1 - (-1)}{2 - 0} = \frac{2}{2} = 1$. The y -intercept is -1, so the line is $y = x - 1$. Check (2, 1): $2 - 1 = 1$. It works.
- Test (0, 5) in the solid line: $x - 1 = -1$, and 5 is greater than -1. So the shading is above the line, giving $y \geq x - 1$.
- The line is solid, so the points on it are included: $y \geq x - 1$ with the non-strict sign.
- Dashed line slope: $\frac{0 - 6}{4 - 1} = \frac{-6}{3} = -2$. Use (4, 0) to find the intercept: $0 = -2(4) + b$, so $0 = -8 + b$ and $b = 8$. The line is $y = -2x + 8$. Check (1, 6): $-2(1) + 8 = 6$. It works.
- Test (0, 5) in the dashed line: $-2(0) + 8 = 8$, and $5 < 8$. So the shading is below the line, giving $y < -2x + 8$.
- The line is dashed, so the points on it are not included: the strict sign $<$ is used.
- The system is $y \geq x - 1$ and $y < -2x + 8$. Option B has $\leq$ (shading below the solid line); option C has $>$ (above the dashed line); option D swaps strict and non-strict. Option A matches.
- So the answer is A.

Answer: (A)

18.

HARD · NO CALCULATOR NEEDED

$$y > 2x - 5$$

$$2y \leq 3x + 11$$

For how many integer values of k is the point $(4, k)$ a solution to the system of inequalities above?

1. Given: the system $y > 2x - 5$ and $2y \leq 3x + 11$, and the point $(4, k)$. We want the number of integer values of k for which $(4, k)$ is a solution.
2. Method: substitute $x = 4$ and $y = k$ into both inequalities to get two inequalities in k , combine them, and count the integers.
3. First inequality: $k > 2(4) - 5$. Since $2(4) = 8$ and $8 - 5 = 3$, we get $k > 3$.
4. Second inequality: $2k \leq 3(4) + 11$. Since $3(4) = 12$ and $12 + 11 = 23$, we get $2k \leq 23$.
5. Divide both sides by 2 (positive, so the sign stays): $k \leq 11.5$.
6. Combine: $3 < k \leq 11.5$.
7. The smallest integer above 3 is 4, and the largest integer not above 11.5 is 11. The integers are 4, 5, 6, 7, 8, 9, 10, 11.
8. Count: $11 - 4 + 1 = 8$.
9. Check the ends: $k = 4$ gives $4 > 3$ true and $2(4) = 8 \leq 23$ true. $k = 11$ gives $11 > 3$ true and $22 \leq 23$ true. $k = 12$ gives $24 \leq 23$, false. $k = 3$ gives $3 > 3$, false.
10. So the answer is 8.

Answer: 8

19.

HARD · CALCULATOR

A 240-milliliter solution is 35% alcohol by volume. How many milliliters of pure water must be added to make a solution that is 28% alcohol by volume?

(A) 84

(B) 300

(C) 45

(D) 60 ✓

1. Given: 240 mL of solution that is 35% alcohol. Let w = the mL of pure water added. The goal is a solution that is 28% alcohol. We want w .
2. Method: adding water does not change the amount of alcohol, so set the original alcohol amount equal to 28% of the new volume.
3. Alcohol in the original solution: $0.35 \times 240 = 70 + 14 = 84$ mL (since $0.35 \times 200 = 70$ and $0.35 \times 40 = 14$).
4. The new volume is $240 + w$ mL, and it must contain 28% alcohol: $84 = 0.28(240 + w)$.
5. Divide both sides by 0.28: $\frac{84}{0.28} = 240 + w$.
6. Compute: $\frac{84}{0.28} = \frac{8400}{28} = 300$. So $300 = 240 + w$.
7. Subtract 240 from both sides: $w = 60$.
8. Check: the new volume is $240 + 60 = 300$ mL, and $0.28 \times 300 = 84$ mL, which is the original alcohol amount.
9. The other choices give a different percent. For example, adding 84 mL gives $\frac{84}{324} \approx 25.9\%$, not 28%.
10. Option D is 60. So the answer is D.

Answer: (D)

20.

HARD · NO CALCULATOR NEEDED

The number of tickets N remaining w weeks after sales began is modeled by $N = 1200 - 45w$. Consider the following statements.

- I. 1,200 tickets were available when sales began.
- II. Each ticket sold reduces the number of remaining tickets by 45.
- III. The model predicts 300 tickets remain after 20 weeks.

Which of the statements are true?

- (A) I only
- (B) I and II only
- (C) I and III only ✓**
- (D) I, II, and III

1. Given: $N = 1200 - 45w$, where N is the tickets remaining after w weeks. We must decide which of statements I, II, III are true.
2. Method: the constant term is the value at $w = 0$, the slope is the change in N per week, and any other claim can be tested by substituting.
3. Statement I: at the start $w = 0$, so $N = 1200 - 45(0) = 1200$. Statement I is true.
4. Statement II: the slope -45 means N drops by 45 for each 1 increase in w . The units are tickets per week.
5. Statement II says 45 tickets are removed per ticket sold. A single ticket sold removes exactly 1 ticket from the remaining count, so this does not match the slope. Statement II is false.
6. Statement III: $N(20) = 1200 - 45(20)$. Since $45 \times 20 = 900$, $N(20) = 1200 - 900 = 300$. Statement III is true.
7. Check III in reverse: set $300 = 1200 - 45w$. Subtract 1200: $-900 = -45w$. Divide by -45 : $w = 20$. This matches.
8. Statements I and III are true and II is false. Option C is 'I and III only'.
9. So the answer is C.

Answer: (C)

21.

HARD · CALCULATOR

The length L , in centimeters, of a metal rod at temperature T degrees Celsius is modeled by $L = 120 + 0.0024T$. The temperature of the rod rises by 90 degrees Fahrenheit. By how many centimeters does the length of the rod increase?

- (A) 0.389
- (B) 0.077
- (C) 0.120 ✓**
- (D) 0.216

1. Given: $L = 120 + 0.0024T$, where L is length in cm and T is temperature in degrees Celsius. The temperature rises by 90 degrees Fahrenheit. We want the increase in L .
2. Method: the model uses Celsius, so convert the 90 degree Fahrenheit change into a Celsius change, then multiply by the slope 0.0024.
3. A change of 1 degree Celsius is a change of 1.8 degrees Fahrenheit, so $\Delta F = 1.8 \Delta C$. The $+32$ in the conversion formula cancels when we look at a change.
4. Substitute $\Delta F = 90$: $90 = 1.8 \Delta C$.
5. Divide both sides by 1.8: $\Delta C = \frac{90}{1.8} = 50$ degrees Celsius.
6. In the model, L changes by the slope times the change in T : $\Delta L = 0.0024 \Delta T$.
7. Substitute $\Delta T = 50$: $\Delta L = 0.0024 \times 50$.
8. Compute: $0.0024 \times 100 = 0.24$, and 50 is half of 100, so $0.0024 \times 50 = 0.12$.
9. Check: $L(T + 50) - L(T) = [120 + 0.0024(T + 50)] - [120 + 0.0024T] = 0.0024(50) = 0.12$.
10. Option C is 0.120. So the answer is C, 0.12 cm.

Answer: (C)

22.

HARD · NO CALCULATOR NEEDED

A triangle has side lengths x , $2x + 1$ and 12, where x is a positive integer. How many different values of x are possible?

(A) 6

(B) 7 ✓

(C) 8

(D) 5

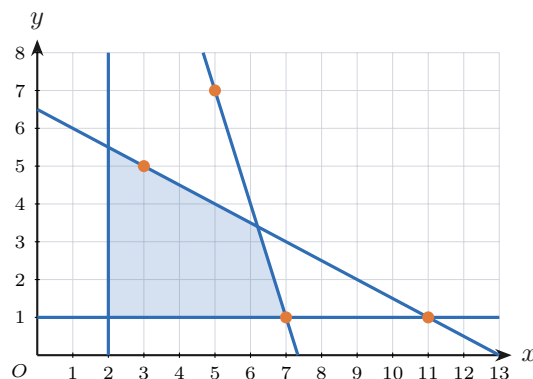
1. Given: a triangle with sides x , $2x + 1$ and 12, where x is a positive integer. We want the number of possible values of x .
2. Method: apply the triangle inequality to all three pairs of sides, solve each, intersect the results, and count the integers.
3. Condition 1: $x + (2x + 1) > 12$. Combine like terms: $3x + 1 > 12$.
4. Subtract 1: $3x > 11$. Divide by 3: $x > \frac{11}{3} = 3.67\dots$. Since x is an integer, $x \geq 4$.
5. Condition 2: $x + 12 > 2x + 1$. Subtract x from both sides: $12 > x + 1$. Subtract 1: $11 > x$, so $x < 11$. Since x is an integer, $x \leq 10$.
6. Condition 3: $(2x + 1) + 12 > x$. Combine: $2x + 13 > x$. Subtract x : $x + 13 > 0$, so $x > -13$. This is true for every positive x .
7. Combine: $4 \leq x \leq 10$.
8. The integers are 4, 5, 6, 7, 8, 9, 10. The count is $10 - 4 + 1 = 7$.
9. Check the ends. $x = 4$: sides 4, 9, 12, and $4 + 9 = 13 > 12$ is true. $x = 10$: sides 10, 21, 12, and $10 + 12 = 22 > 21$ is true.
10. Check just outside. $x = 3$: sides 3, 7, 12, and $3 + 7 = 10 > 12$ is false. $x = 11$: sides 11, 23, 12, and $11 + 12 = 23 > 23$ is false.
11. Option B is 7. So the answer is B, 7.

Answer: (B)

23.

HARD · NO CALCULATOR NEEDED

A bakery makes x loaves of rye and y loaves of sourdough each day. The shaded region shows all combinations (x, y) the bakery can produce, with the boundaries shown as solid lines. On a day when the bakery makes 4 loaves of rye, what is the greatest number of loaves of sourdough it can make?



1. Given (from the figure): one boundary passes through $(3, 5)$ and $(11, 1)$; another passes through $(5, 7)$ and $(7, 1)$; the region lies on the origin side of both, with solid boundaries. With $x = 4$ loaves of rye, we want the greatest number of sourdough loaves y .
2. Method: find each boundary line, substitute $x = 4$ into each to get an upper limit for y , and use the smaller limit.
3. First boundary slope: $\frac{1 - 5}{11 - 3} = \frac{-4}{8} = -\frac{1}{2}$.
4. Use point-slope form with $(3, 5)$: $y - 5 = -\frac{1}{2}(x - 3)$. Distribute: $y - 5 = -\frac{1}{2}x + \frac{3}{2}$. Add 5: $y = -\frac{1}{2}x + \frac{13}{2}$.
5. Multiply both sides by 2: $2y = -x + 13$. Add x : $x + 2y = 13$. Check $(11, 1)$: $11 + 2(1) = 13$. It works. The region is $x + 2y \leq 13$.
6. Second boundary slope: $\frac{1 - 7}{7 - 5} = \frac{-6}{2} = -3$.
7. Point-slope with $(7, 1)$: $y - 1 = -3(x - 7)$. Distribute: $y - 1 = -3x + 21$. Add 1: $y = -3x + 22$, which is $3x + y = 22$. Check $(5, 7)$: $3(5) + 7 = 22$. It works. The region is $3x + y \leq 22$.
8. Substitute $x = 4$ into the first boundary: $4 + 2y \leq 13$. Subtract 4: $2y \leq 9$. Divide by 2: $y \leq 4.5$.
9. Substitute $x = 4$ into the second boundary: $3(4) + y \leq 22$, so $12 + y \leq 22$. Subtract 12: $y \leq 10$.
10. Both limits must hold, so use the smaller one: $y \leq 4.5$.
11. Loaves are whole numbers, so the greatest value is $y = 4$.
12. Check $(4, 4)$: $4 + 2(4) = 12 \leq 13$ true, and $3(4) + 4 = 16 \leq 22$ true. Check $(4, 5)$: $4 + 2(5) = 14 \leq 13$ is false. So the answer is 4.

Answer: 4

24.

SAT CHALLENGE · CALCULATOR

Plan A costs \$40 per month plus \$3 per gigabyte (GB) of data. Plan B costs \$25 per month for the first 4 GB plus \$8 for each GB beyond 4 GB, and then an 8% fee is added to the monthly total. For whole-number usage $g \geq 4$ GB, what is the least value of g for which Plan B costs more than Plan A?

- Given: Plan A is \$40 per month plus \$3 per GB. Plan B is \$25 for the first 4 GB plus \$8 for each GB beyond 4, and then an 8% fee is added to that total. Let g = GB used ($g \geq 4$, whole number). We want the least g for which B costs more than A.
- Method: write each plan's cost as a function of g , set up $B > A$, solve, and round up to a whole number.
- Plan A: $A = 40 + 3g$.
- Plan B before the fee: $25 + 8(g - 4)$. Distribute: $25 + 8g - 32$. Combine constants: $8g - 7$.
- Add the 8% fee by multiplying by 1.08: $B = 1.08(8g - 7)$.
- Distribute: $1.08 \times 8 = 8.64$ and $1.08 \times 7 = 7.56$, so $B = 8.64g - 7.56$.
- Set up $B > A$: $8.64g - 7.56 > 40 + 3g$.
- Subtract $3g$ from both sides: $5.64g - 7.56 > 40$.
- Add 7.56 to both sides: $5.64g > 47.56$.
- Divide both sides by 5.64 (positive, so the sign stays): $g > \frac{47.56}{5.64} \approx 8.43$.
- g is a whole number greater than 8.43, so the least value is 9.
- Check $g = 8$: $A = 40 + 3(8) = 64$; $B = 1.08(8 \cdot 8 - 7) = 1.08(57) = 61.56$. Since $61.56 > 64$ is false, B is not more expensive.
- Check $g = 9$: $A = 40 + 3(9) = 67$; $B = 1.08(8 \cdot 9 - 7) = 1.08(65) = 70.20$. Since $70.20 > 67$ is true, B costs more.
- So the answer is 9.

Answer: 9**25.**

SAT CHALLENGE · NO CALCULATOR NEEDED

$$y \geq 3x + a$$

$$2y < 6x - 8$$

In the system of inequalities above, a is a constant. For the least integer value of a for which the system has no solution, what is the value of a ?

- Given: $y \geq 3x + a$ and $2y < 6x - 8$, where a is a constant. We want the least integer a for which the system has no solution.
- Method: write both inequalities as y compared with $3x + \text{constant}$. Both lines have slope 3, so the question is whether there is room for y between them.
- Divide both sides of the second inequality by 2 (positive, so the sign stays): $y < 3x - 4$.
- The system is now $3x + a \leq y < 3x - 4$.
- For a given x , some y exists only if the lower bound $3x + a$ is below the upper bound $3x - 4$. The upper bound is strict, so the two bounds cannot be equal either.
- So we need $3x + a < 3x - 4$. Subtract $3x$ from both sides: $a < -4$.
- Therefore solutions exist exactly when $a < -4$, and there is no solution exactly when $a \geq -4$.
- The integers with no solution are $-4, -3, -2, \dots$. The least is -4 .
- Check $a = -4$: the system is $y \geq 3x - 4$ and $y < 3x - 4$. No y can be both at least and less than the same value, so there is no solution.
- Check $a = -5$: the system is $3x - 5 \leq y < 3x - 4$. Take $x = 0$: $-5 \leq y < -4$, and $y = -5$ works. So $a = -5$ has a solution and is not the answer.
- So the answer is $a = -4$.

Answer: -4

26.

SAT CHALLENGE · DESMOS

In the xy -plane, the region defined by $x \geq 0$, $y \geq 0$, $x \leq 5$, $3x + 2y \leq 18$ and $x + 2y \leq 10$ is a polygon. What is the area of the polygon?

1. Given: the region $x \geq 0$, $y \geq 0$, $x \leq 5$, $3x + 2y \leq 18$, $x + 2y \leq 10$. We want the area of this polygon.
2. Method: find the corner points by intersecting boundary lines, keep those that satisfy every inequality, then use the shoelace formula for the area.
3. Intersect $3x + 2y = 18$ and $x + 2y = 10$. Subtract the second equation from the first: $(3x + 2y) - (x + 2y) = 18 - 10$, so $2x = 8$ and $x = 4$.
4. Substitute $x = 4$ into $x + 2y = 10$: $4 + 2y = 10$. Subtract 4: $2y = 6$. Divide by 2: $y = 3$. The point is $(4, 3)$.
5. Check $(4, 3)$ in all the inequalities: $4 \leq 5$ true; $3(4) + 2(3) = 18 \leq 18$ true; $4 + 2(3) = 10 \leq 10$ true. It is a corner.
6. Intersect $x = 5$ with $3x + 2y = 18$: $15 + 2y = 18$. Subtract 15: $2y = 3$. Divide by 2: $y = 1.5$. The point is $(5, 1.5)$. Check $x + 2y = 5 + 3 = 8 \leq 10$, true. It is a corner.
7. Intersect $x = 0$ with $x + 2y = 10$: $2y = 10$, so $y = 5$. The point is $(0, 5)$. Check $3(0) + 2(5) = 10 \leq 18$, true. It is a corner.
8. On the axes: $(0, 0)$ and $(5, 0)$ satisfy every inequality ($3(5) = 15 \leq 18$ and $5 \leq 10$). The lines $3x + 2y = 18$ and $x + 2y = 10$ meet $y = 0$ at $x = 6$ and $x = 10$, which are beyond $x = 5$, so they are not corners.
9. The corners in order around the polygon: $(0, 0)$, $(5, 0)$, $(5, 1.5)$, $(4, 3)$, $(0, 5)$.
10. Shoelace terms $x_i y_{i+1} - x_{i+1} y_i$: $(0, 0) \rightarrow (5, 0)$ gives $0 \cdot 0 - 5 \cdot 0 = 0$; $(5, 0) \rightarrow (5, 1.5)$ gives $5(1.5) - 5(0) = 7.5$; $(5, 1.5) \rightarrow (4, 3)$ gives $5(3) - 4(1.5) = 15 - 6 = 9$; $(4, 3) \rightarrow (0, 5)$ gives $4(5) - 0(3) = 20$; $(0, 5) \rightarrow (0, 0)$ gives 0.
11. Add the terms: $0 + 7.5 + 9 + 20 + 0 = 36.5$.
12. Area = $\frac{36.5}{2} = 18.25$.
13. Check by splitting into two strips: for $0 \leq x \leq 4$ the top edge goes from height 5 to height 3, area $\frac{5+3}{2} \cdot 4 = 16$. For $4 \leq x \leq 5$ the top edge goes from 3 to 1.5, area $\frac{3+1.5}{2} \cdot 1 = 2.25$. Total $16 + 2.25 = 18.25$.
14. So the answer is 18.25.

Answer: 18.25**27.**

SAT CHALLENGE · CALCULATOR

A bakery's daily cost is \$120 plus \$0.75 per loaf. On one day it sells 500 loaves: the first 100 at a price of p dollars each and the remaining loaves at 80% of that price. What is the least price p , in dollars, that is a whole number of cents and gives a profit of at least \$200 that day?

1. Given: daily cost is \$120 plus \$0.75 per loaf. 500 loaves are sold: the first 100 at price p and the other 400 at 80% of p . The profit must be at least \$200. Let p = the price in dollars. We want the least p in whole cents.
2. Method: profit = revenue - cost. Write both in terms of p , then solve the inequality (profit) ≥ 200 .
3. Cost: $120 + 0.75(500) = 120 + 375 = 495$ dollars.
4. The remaining loaves: $500 - 100 = 400$, each sold at $0.80p$.
5. Revenue: $100p + 400(0.80p)$. Since $400 \times 0.80 = 320$, revenue = $100p + 320p = 420p$.
6. Profit: $420p - 495$. Require $420p - 495 \geq 200$.
7. Add 495 to both sides: $420p \geq 695$.
8. Divide both sides by 420 (positive, so the sign stays): $p \geq \frac{695}{420} = 1.6547 \dots$ (since $420 \times 1.65 = 693$).
9. Prices are in whole cents. The cent values just below and above the bound are 1.65 and 1.66. The price must be at least 1.6547..., so 1.65 is too small and the least allowed price is 1.66.
10. Check $p = 1.65$: revenue $420(1.65) = 693$, profit $693 - 495 = 198$, and $198 \geq 200$ is false.
11. Check $p = 1.66$: revenue $420(1.66) = 697.20$, profit $697.20 - 495 = 202.20$, and $202.20 \geq 200$ is true.
12. So the answer is \$1.66.

Answer: 1.66

28.

SAT CHALLENGE · CALCULATOR

A jeweler melts together 12 kilograms of an alloy that is 25% silver, some amount of an alloy that is 60% silver, and some amount of pure copper (which contains no silver) to make 40 kilograms of an alloy that is 36% silver. How many kilograms of pure copper were used?

(A) 14.4

(B) 19

(C) 28

(D) 9 ✓

1. Given: 12 kg of an alloy that is 25% silver, x kg of an alloy that is 60% silver, and y kg of pure copper (0% silver) make 40 kg of an alloy that is 36% silver. We want y .
2. Method: write two equations, one for total mass and one for the amount of silver, and solve them.
3. Mass equation: $12 + x + y = 40$. Subtract 12 from both sides: $x + y = 28$.
4. Silver in each part: $0.25(12) = 3$ kg from the first alloy, $0.60x$ kg from the second, and 0 from the copper.
5. Silver needed in the final alloy: $0.36(40) = 14.4$ kg.
6. Silver equation: $3 + 0.6x = 14.4$.
7. Subtract 3 from both sides: $0.6x = 11.4$.
8. Divide both sides by 0.6: $x = \frac{11.4}{0.6} = \frac{114}{6} = 19$.
9. Substitute into $x + y = 28$: $19 + y = 28$. Subtract 19: $y = 9$.
10. Check the mass: $12 + 19 + 9 = 40$. Check the silver: $3 + 0.6(19) = 3 + 11.4 = 14.4$, and $\frac{14.4}{40} = 0.36 = 36\%$.
11. The question asks for the copper, which is $y = 9$ kg. This is option D.
12. So the answer is D, 9.

Answer: (D)**29.**

SAT CHALLENGE · NO CALCULATOR NEEDED

A triangle has side lengths $2x$, $x + 6$ and $18 - x$, where x is an integer. What is the sum of all possible perimeters of the triangle?

1. Given: a triangle with sides $2x$, $x + 6$ and $18 - x$, where x is an integer. We want the sum of all possible perimeters.
2. Method: find the integers x that give a real triangle (positive sides and the triangle inequality), write the perimeter as a function of x , and add the perimeters.
3. Positive sides: $2x > 0$ gives $x > 0$; $18 - x > 0$ gives (add x) $18 > x$, so $x < 18$. The side $x + 6 > 0$ holds when $x > 0$.
4. Inequality 1: $2x + (x + 6) > 18 - x$. Combine: $3x + 6 > 18 - x$. Add x : $4x + 6 > 18$. Subtract 6: $4x > 12$. Divide by 4: $x > 3$.
5. Inequality 2: $2x + (18 - x) > x + 6$. The left side is $x + 18$, so $x + 18 > x + 6$. Subtract x : $18 > 6$, which is always true.
6. Inequality 3: $(x + 6) + (18 - x) > 2x$. The left side is 24, so $24 > 2x$. Divide by 2: $x < 12$.
7. Combine all conditions: $3 < x < 12$, so $x = 4, 5, 6, 7, 8, 9, 10, 11$. That is 8 values.
8. Perimeter: $2x + (x + 6) + (18 - x) = 2x + x - x + 6 + 18 = 2x + 24$.
9. The perimeters are 32, 34, 36, 38, 40, 42, 44, 46 for $x = 4$ up to $x = 11$.
10. Sum of an arithmetic list: 8 terms, first 32, last 46, so the sum is $8 \times \frac{32 + 46}{2} = 8 \times 39 = 312$.
11. Check another way: $4 + 5 + \dots + 11 = 8 \times \frac{4 + 11}{2} = 60$, so $\sum(2x + 24) = 2(60) + 8(24) = 120 + 192 = 312$.
12. Check the end triangles: $x = 4$ gives sides 8, 10, 14 and $8 + 10 = 18 > 14$; $x = 11$ gives sides 22, 17, 7 and $17 + 7 = 24 > 22$. Both are valid.
13. So the answer is 312.

Answer: 312

30.

SAT CHALLENGE · DESMOS

Candle A is 24 centimeters tall and burns down at a constant rate of 2 centimeters per hour. Candle B is 30 centimeters tall and burns down at a constant rate of 3.5 centimeters per hour. Both are lit at the same time, and each stays at 0 centimeters once it has burned out. For how many whole numbers of hours t , with $t \geq 0$, is candle A at least 3 centimeters taller than candle B?

(A) 5 ✓

(B) 6

(C) 3

(D) 4

- Given: candle A starts at 24 cm and burns 2 cm per hour; candle B starts at 30 cm and burns 3.5 cm per hour; each stays at 0 once burned out. We want how many whole numbers $t \geq 0$ make A at least 3 cm taller than B, that is $A - B \geq 3$.
- Heights: $A(t) = 24 - 2t$ until it burns out, and $B(t) = 30 - 3.5t$ until it burns out.
- Burn-out times: A at $\frac{24}{2} = 12$ hours; B at $\frac{30}{3.5} \approx 8.57$ hours. So for whole numbers, B is still burning for $t \leq 8$ and is out for $t \geq 9$.
- Method: split into cases, since the formula for B changes at $t = 9$.
- Case 1, $0 \leq t \leq 8$ (both burning): $A - B = (24 - 2t) - (30 - 3.5t)$. Distribute the minus sign: $24 - 2t - 30 + 3.5t$. Combine like terms: $1.5t - 6$.
- Require $1.5t - 6 \geq 3$. Add 6 to both sides: $1.5t \geq 9$. Divide by 1.5: $t \geq 6$.
- With $t \leq 8$, the values are $t = 6, 7, 8$.
- Check the edge: $t = 6$ gives $A = 12$, $B = 9$, difference 3, which works. $t = 5$ gives $A = 14$, $B = 12.5$, difference 1.5, which does not.
- Case 2, $9 \leq t \leq 11$ (B is out, so $B = 0$, and A still burns): $A - B = 24 - 2t$. Require $24 - 2t \geq 3$.
- Subtract 24 from both sides: $-2t \geq -21$. Divide by -2 , which is negative, so the inequality sign flips: $t \leq 10.5$.
- With $t \geq 9$ and whole numbers, the values are $t = 9, 10$. Check $t = 11$: $24 - 22 = 2$, and $2 \geq 3$ is false.
- Case 3, $t \geq 12$: both candles are at 0, the difference is 0, and $0 \geq 3$ is false.
- All values: $t = 6, 7, 8, 9, 10$, which is 5 values. Option A is 5. So the answer is A.

Answer: (A)

31.

SAT CHALLENGE · CALCULATOR

A utility charges a \$15 service fee each month. Usage is billed at \$2.40 per unit for the first 20 units and \$3.60 for each unit beyond 20. A 10% fuel surcharge is added to the usage charge only (not to the service fee). Then 5% tax is applied to the sum of the service fee and the surcharged usage charge. For $u > 20$ units, which function gives the total monthly bill $B(u)$, in dollars?

(A) $B(u) = 3.78u - 9.45$ (B) $B(u) = 4.158u - 11.97$ ✓(C) $B(u) = 4.158u - 10.395$ (D) $B(u) = 4.158u - 12.72$

- Given: a \$15 monthly service fee; usage billed at \$2.40 per unit for the first 20 units and \$3.60 per unit beyond 20; a 10% surcharge on usage only; then 5% tax on (service fee + surcharged usage). Here $u > 20$. We want $B(u)$.
- Method: build the bill one stage at a time, in the order the problem states.
- Usage for the first 20 units: $2.40 \times 20 = 48$ dollars.
- Usage beyond 20: there are $u - 20$ units at \$3.60, so $3.6(u - 20) = 3.6u - 72$.
- Total usage charge: $48 + 3.6u - 72$. Combine constants: $3.6u - 24$.
- The 10% surcharge applies to usage only. Multiply by 1.10: $1.1(3.6u - 24) = 3.96u - 26.4$ (since $1.1 \times 3.6 = 3.96$ and $1.1 \times 24 = 26.4$).
- Add the service fee, which gets no surcharge: $15 + 3.96u - 26.4$. Combine constants: $3.96u - 11.4$.
- The 5% tax applies to this whole sum. Multiply by 1.05: $B = 1.05(3.96u - 11.4)$.
- Distribute: $1.05 \times 3.96 = 4.158$ and $1.05 \times 11.4 = 11.97$. So $B(u) = 4.158u - 11.97$.
- Check with $u = 30$ directly: usage $48 + 3.6(10) = 84$; with surcharge $84 \times 1.1 = 92.40$; add the fee 107.40; with tax $107.40 \times 1.05 = 112.77$.
- Check with the formula: $4.158(30) - 11.97 = 124.74 - 11.97 = 112.77$. The two agree.
- Option B is $B(u) = 4.158u - 11.97$. So the answer is B.

Answer: (B)

32.

SAT CHALLENGE · CALCULATOR

A utility computes a customer's monthly water bill B , in dollars, for u units of water in this order: a service fee of \$18 plus \$2.40 for each unit used; then a one-time credit of \$12 is subtracted; then a 5% surcharge is added to the amount at that point; and finally a fixed \$3.50 fee is added. Which of the following defines B for u units?

(A) $B = 2.52u + 9.975$

(B) $B = 2.52u + 22.4$

(C) $B = 2.52u + 9.8$ ✓

(D) $B = 2.52u + 10.4$

- Given: the bill is built in this order: fee \$18 plus \$2.40 per unit; subtract a \$12 credit; add a 5% surcharge to the amount at that point; then add a fixed \$3.50. Let u = units used and B = the bill. We want B in terms of u .
- Method: apply each stage in the stated order, simplifying as we go.
- After the fee and usage: $18 + 2.4u$.
- After the credit, subtract 12: $18 + 2.4u - 12$. Combine constants: $6 + 2.4u$.
- A 5% surcharge on this amount multiplies it by 1.05: $1.05(6 + 2.4u)$.
- Distribute: $1.05 \times 6 = 6.3$ and $1.05 \times 2.4u = 2.52u$. So the amount is $6.3 + 2.52u$.
- The fixed fee comes last, after the surcharge, so it is just added: $6.3 + 2.52u + 3.5$.
- Combine constants: $6.3 + 3.5 = 9.8$. So $B = 2.52u + 9.8$.
- Check with $u = 10$ step by step: $18 + 24 = 42$; $42 - 12 = 30$; $30 \times 1.05 = 31.5$; $31.5 + 3.5 = 35$. Formula: $2.52(10) + 9.8 = 25.2 + 9.8 = 35$. They agree.
- Check with $u = 0$: $18 - 12 = 6$; $6 \times 1.05 = 6.3$; $6.3 + 3.5 = 9.8$. Formula: $2.52(0) + 9.8 = 9.8$. They agree.
- Only option C has slope 2.52 and constant 9.8. So the answer is C.

Answer: (C)

33.

SAT CHALLENGE · CALCULATOR

A ride service charges a fixed pickup fee plus a constant rate for each mile traveled. A 5-mile ride costs \$14.50 and a 12-mile ride costs \$29.20. The service then raises its per-mile rate by 20% and leaves the pickup fee unchanged. Using 1 mile = 1.6 kilometers, what will a 20-kilometer ride cost after the change?

(A) \$35.50 ✓

(B) \$36.30

(C) \$54.40

(D) \$84.64

- Given: fare = fixed fee + rate $\times$ miles. A 5-mile ride costs \$14.50; a 12-mile ride costs \$29.20. Then the per-mile rate rises 20% and the fee stays. Let f = fee and r = rate per mile. We want the cost of a 20-km ride after the change, with 1 mile = 1.6 km.
- Method: find r and f from the two rides (a linear model), apply the 20% increase to r only, convert kilometers to miles, and evaluate.
- Write the two equations: $f + 5r = 14.50$ and $f + 12r = 29.20$.
- Subtract the first from the second so f cancels: $7r = 14.70$.
- Divide by 7: $r = 2.10$ dollars per mile.
- Substitute into the first equation: $f + 5(2.10) = 14.50$, so $f + 10.50 = 14.50$. Subtract 10.50: $f = 4.00$ dollars.
- Check with the second ride: $4 + 12(2.10) = 4 + 25.20 = 29.20$. It matches.
- New rate: $2.10 \times 1.20 = 2.52$ dollars per mile. The fee stays \$4.00.
- Convert the distance: $20 \text{ km} \div 1.6 \text{ km per mile} = 12.5 \text{ miles}$.
- New cost: $4.00 + 2.52(12.5)$. Since $2.52 \times 12.5 = \frac{252}{8} = 31.5$, the cost is $4.00 + 31.50 = 35.50$.
- Option A is \$35.50, which matches.
- So the answer is A, \$35.50.

Answer: (A)

34.

SAT CHALLENGE · DESMOS

A coffee shop blends Ethiopian beans, which cost \$18 per kilogram, with Colombian beans, which cost \$11 per 500 grams. The blend has a total mass of 36 kilograms, and its cost works out to \$20.50 per kilogram. How many kilograms of Ethiopian beans are in the blend?

(A) 22.5

(B) 18

(C) 27

(D) 13.5 ✓

- Given: Ethiopian beans cost \$18 per kilogram; Colombian beans cost \$11 per 500 grams; the blend is 36 kg at \$20.50 per kilogram. Let x = kg of Ethiopian beans and y = kg of Colombian beans. We want x .
- Put both prices in the same unit. 500 g = 0.5 kg, so Colombian beans cost $\frac{11}{0.5} = 22$ dollars per kilogram.
- Method: write a mass equation and a cost equation and solve by substitution.
- Mass: $x + y = 36$.
- Cost: the blend costs $20.50 \times 36 = 738$ dollars in total (since $20 \times 36 = 720$ and $0.5 \times 36 = 18$). So $18x + 22y = 738$.
- From the mass equation, subtract y : $x = 36 - y$.
- Substitute into the cost equation: $18(36 - y) + 22y = 738$. Distribute: $648 - 18y + 22y = 738$.
- Combine like terms: $648 + 4y = 738$. Subtract 648: $4y = 90$. Divide by 4: $y = 22.5$.
- Then $x = 36 - 22.5 = 13.5$.
- Check the mass: $13.5 + 22.5 = 36$. Check the cost: $18(13.5) = 243$ and $22(22.5) = 495$, and $243 + 495 = 738$.
- The question asks for the Ethiopian beans, which is $x = 13.5$ kg. This is option D.
- So the answer is D, 13.5.

Answer: (D)

35.

SAT CHALLENGE · CALCULATOR

Plan P costs \$25 per month for up to 4 gigabytes (GB) of data, plus \$6 for each additional GB. Plan Q costs \$14 per month for up to 2 GB of data, plus \$0.008 for each additional megabyte (MB). Charges for additional data are prorated, and 1 GB = 1000 MB. For what amount of data, in MB, do the two plans cost the same?

- Given: Plan P is \$25 for up to 4 GB, plus \$6 per additional GB. Plan Q is \$14 for up to 2 GB, plus \$0.008 per additional MB. Extra data is prorated and 1 GB = 1000 MB. We want the amount of data, in MB, at which the plans cost the same.
- Method: use one unit (GB), write each plan's cost on each range of usage, find where the costs are equal, then convert to MB.
- Plan Q's extra charge in GB: $0.008 \times 1000 = 8$ dollars per GB.
- Let g = GB used. Plan P: 25 for $g \leq 4$, and $25 + 6(g - 4)$ for $g > 4$. Plan Q: 14 for $g \leq 2$, and $14 + 8(g - 2)$ for $g > 2$.
- Range $g \leq 2$: P costs 25 and Q costs 14. They are not equal, so there is no solution here.
- Range $2 < g \leq 4$: P costs 25 and Q costs $14 + 8(g - 2)$. Set them equal: $14 + 8(g - 2) = 25$.
- Subtract 14 from both sides: $8(g - 2) = 11$. Divide by 8: $g - 2 = 1.375$. Add 2: $g = 3.375$.
- Since $2 < 3.375 \leq 4$, this value is in the range.
- Range $g > 4$: P costs $25 + 6(g - 4) = 25 + 6g - 24 = 6g + 1$ and Q costs $14 + 8(g - 2) = 14 + 8g - 16 = 8g - 2$. Set them equal: $6g + 1 = 8g - 2$.
- Subtract $6g$: $1 = 2g - 2$. Add 2: $3 = 2g$. Divide by 2: $g = 1.5$. This is not greater than 4, so there is no solution in this range.
- The only solution is $g = 3.375$ GB. Convert to MB: $3.375 \times 1000 = 3375$ MB.
- Check: Q at 3.375 GB costs $14 + 8(1.375) = 14 + 11 = 25$, which equals P's flat \$25. So the answer is 3375.

Answer: 3375

36.

SAT CHALLENGE · CALCULATOR

Plan A charges a one-time enrollment fee of \$85 plus \$18.50 per month. Plan B has no enrollment fee and charges \$28 per month. For a whole number of months m , which of the following describes all values of m for which Plan A costs strictly less than Plan B in total?

- (A) $m \geq 10$ (B) $m \geq 8$
 (C) $m \geq 9$ ✓ (D) $m \leq 8$

- Given: Plan A has a one-time fee of \$85 plus \$18.50 per month. Plan B has no fee and costs \$28 per month. Let m = the number of months (a whole number). We want all m for which A costs strictly less than B.
- Write the total costs: Plan A: $85 + 18.5m$. Plan B: $28m$.
- Method: set up the inequality $A < B$ and solve for m .
- Write the inequality: $85 + 18.5m < 28m$.
- Subtract $18.5m$ from both sides: $85 < 9.5m$, because $28 - 18.5 = 9.5$.
- Divide both sides by 9.5 (positive, so the sign stays): $\frac{85}{9.5} < m$, which is $m > \frac{170}{19} \approx 8.95$.
- The whole numbers greater than 8.95 start at 9, so $m \geq 9$.
- Check $m = 8$: A costs $85 + 18.5(8) = 85 + 148 = 233$ and B costs $28(8) = 224$. Since $233 < 224$ is false, A is not cheaper.
- Check $m = 9$: A costs $85 + 18.5(9) = 85 + 166.5 = 251.5$ and B costs $28(9) = 252$. Since $251.5 < 252$ is true, A is cheaper.
- Check $m = 10$: A costs $85 + 185 = 270$ and B costs 280. A is cheaper again, and the gap grows by 9.50 each month.
- Compare the choices: option A ($m \geq 10$) leaves out $m = 9$; option B ($m \geq 8$) includes $m = 8$; option D ($m \leq 8$) points the wrong way. Option C matches.
- So the answer is C, $m \geq 9$.

Answer: (C)

37.

SAT CHALLENGE · DESMOS

$$x \geq 0, \quad y \geq 0, \quad 2x + y \leq 10, \quad x + 3y \leq 15, \quad 3x + 2y > k$$

In the system of inequalities above, k is a constant. What is the least value of k for which the system has no solution (x, y) ?

- Given: $x \geq 0$, $y \geq 0$, $2x + y \leq 10$, $x + 3y \leq 15$, and $3x + 2y > k$. We want the least k for which the system has no solution.
- Method: the first four inequalities describe a polygon. The system has a solution if some point of the polygon has $3x + 2y > k$. So we need the greatest value of $3x + 2y$ on the polygon, which occurs at a corner.
- Corner at the origin: $(0, 0)$.
- Corner on the x -axis: set $y = 0$ in $2x + y = 10$ to get $2x = 10$, $x = 5$. The point $(5, 0)$ satisfies $x + 3y = 5 \leq 15$, so it is a corner.
- Corner on the y -axis: set $x = 0$ in $x + 3y = 15$ to get $3y = 15$, $y = 5$. The point $(0, 5)$ satisfies $2(0) + 5 = 5 \leq 10$, so it is a corner.
- Intersection of $2x + y = 10$ and $x + 3y = 15$: from the first, subtract $2x$: $y = 10 - 2x$.
- Substitute into the second: $x + 3(10 - 2x) = 15$. Distribute: $x + 30 - 6x = 15$. Combine: $-5x + 30 = 15$. Subtract 30: $-5x = -15$. Divide by -5 : $x = 3$. Then $y = 10 - 2(3) = 4$. The point is $(3, 4)$.
- Check $(3, 4)$: $2(3) + 4 = 10 \leq 10$ and $3 + 3(4) = 15 \leq 15$. It is a corner.
- Evaluate $3x + 2y$ at the corners: $(0, 0)$ gives 0; $(5, 0)$ gives 15; $(0, 5)$ gives 10; $(3, 4)$ gives $9 + 8 = 17$.
- A linear expression reaches its greatest value over a polygon at a corner, so the greatest value of $3x + 2y$ is 17.
- If $k < 17$, the point $(3, 4)$ has $3x + 2y = 17 > k$, so the system has a solution.
- If $k = 17$, we would need $3x + 2y > 17$, but every point of the polygon has $3x + 2y \leq 17$. So there is no solution. The same holds for every $k > 17$.
- Check $k = 16.9$: $(3, 4)$ gives $17 > 16.9$, which is true, so a solution exists and 16.9 is not the answer.
- So the answer is 17.

Answer: 17